

Launch Lens

A voice companion for exploring past and upcoming rocket launches.

The Idea

Ask “*What’s launching this week, and where is it going?*” Launch Lens shows past and upcoming launches, booster landings, and orbit ground tracks on a map, and lets you explore them by voice. Grok Imagine generates mission-patch-style artwork for each launch.

Track Fit

Built with Cursor	Data fetching, map, and UI built in Cursor.
Grok Voice / Imagine	Voice Q&A and Imagine mission patches. Uses both APIs.
Real space data	Launch Library 2 (The Space Devs) launch data; orbital data for ground tracks.
Navigation theme	Weak. Ground tracks and booster-recovery ship routes help a little.
Also eligible	Software, Design, People’s Choice.

Features

MVP

- Timeline of upcoming and recent launches with rocket, mission, and launch site.
- Map with launch site and ground track for a selected mission.
- Voice Q&A about launches using Grok.

Stretch

- Grok Imagine mission patch for each launch.
- Booster landing and recovery-ship locations.
- “Can I see it from here?” visibility check for your location.

Tech Stack and Data

- Data: Launch Library 2 API; CelesTrak for orbits once in space.
- Frontend: React with Leaflet or Mapbox.
- AI: Grok Voice and Grok Imagine.

Limitations and Risks

- **Weak theme fit.** Hard to argue it changes how we navigate in the next 100 years, which hurts in the Big Red track.
Mitigation: Lean into ground tracks and “where will it fly over”, or accept that it targets the

SpaceX track only.

- **Least original.** Many launch tracker apps and sites already exist (Next Spaceflight, Rocket-Launch.live).

Mitigation: Voice-first exploration and generated patches are the only differentiators; that may not be enough.

- **API rate limits.** The free Launch Library 2 tier allows only a small number of requests per hour.

Mitigation: Fetch once, cache to a JSON file, and serve from cache. Use their development server while building.

- **Stale community APIs.** The popular community SpaceX API is no longer maintained, so recent launches may be missing.

Mitigation: Use Launch Library 2 as the main source.

- **Mission patches can mislead.** Generated patches could be confused with official mission artwork.

Mitigation: Label them as AI-generated fan art.

Scorecard

Demo-able	4/5	Easy, reliable screen demo.
Buildable in 24h	5/5	Simplest of the six.
Navigation theme	2/5	Weak.
Wow factor	2/5	Feels familiar.
Technical risk (5 = riskiest)	1/5	Low.

Verdict

Fallback only. Choose it for a beginner-heavy team or if another idea collapses on Saturday. Unlikely to stand out on originality or theme.